In [1]:

```
# import library
from google.colab import files
uploaded = files.upload()
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

Choose files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving Features data set.csv to Features data set.csv
Saving sales data-set.csv to sales data-set.csv
Saving stores data-set.csv to stores data-set.csv
```

#introduction:

This project analyzes data to understand key factors influencing revenue. The analysis combines multiple datasets including sales, store information, and external features such as temperature, fuel price, and unemployment.

In [2]:

```
# load the data set
sales_data = pd.read_csv("sales data-set.csv")
features = pd.read_csv("Features data set.csv")
store = pd.read_csv("stores data-set.csv")
```

#Data preparation:

The datasets were first cleaned individually to ensure consistency. Missing values in MarkDown columns were replaced with 0, as they represent weeks with no promotions. Missing values in CPI and Unemployment were filled using mean values. The datasets were then merged using common columns such as Store and Date to create a unified dataset for analysis.

In [3]:

```
# checking for missing values in sales data-set.csv
sales_data.isnull().sum()
```

Out [3]:

| | 0 |
|---|---|
| **Store** | 0 |
| **Dept** | 0 |
| **Date** | 0 |
| **Weekly_Sales** | 0 |
| **IsHoliday** | 0 |

**dtype:** int64

In [4]:

```
#checking the missing value in features data-set.csv
features.isnull().sum()
```

Out [4]:

| | 0 |
|---|---|
| **Store** | 0 |
| **Date** | 0 |
| **Temperature** | 0 |
| **Fuel_Price** | 0 |
| **MarkDown1** | 4158 |
| **MarkDown2** | 5269 |
| **MarkDown3** | 4577 |
| **MarkDown4** | 4726 |
| **MarkDown5** | 4140 |
| **CPI** | 585 |
| **Unemployment** | 585 |
| **IsHoliday** | 0 |

**dtype:** int64

In [5]:

```
# fill missing value in markdown colums with 0 to aoid losing huge data set and ensure good analysis
cols = ["MarkDown1","MarkDown2","MarkDown3","MarkDown4","MarkDown5"]
```

```
features[cols] = features[cols].fillna(0)

#fill null values in CPI an Unemployment column with their mean values
features["CPI"].fillna(features["CPI"].mean(), inplace=True)
features["Unemployment"].fillna(features["Unemployment"].mean(), inplace=True)
```

```
/tmp/ipykernel_311/2733193788.py:6: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained
assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting
values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] =
df[col].method(value) instead, to perform the operation inplace on the original object.


  features["CPI"].fillna(features["CPI"].mean(), inplace=True)
/tmp/ipykernel_311/2733193788.py:7: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained
assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting
values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] =
df[col].method(value) instead, to perform the operation inplace on the original object.


  features["Unemployment"].fillna(features["Unemployment"].mean(), inplace=True)
```

In [6]:

```
#cross check the missing valus
features.isnull().sum()
```

Out [6]:

| | 0 |
|---|---|
| **Store** | 0 |
| **Date** | 0 |
| **Temperature** | 0 |
| **Fuel_Price** | 0 |
| **MarkDown1** | 0 |
| **MarkDown2** | 0 |
| **MarkDown3** | 0 |
| **MarkDown4** | 0 |
| **MarkDown5** | 0 |
| **CPI** | 0 |
| **Unemployment** | 0 |
| **IsHoliday** | 0 |

**dtype:** int64

In [7]:

```
# checking missing values in stores data-set.csv
store.isnull().sum()
```

Out [7]:

| | 0 |
|---|---|
| **Store** | 0 |
| **Type** | 0 |
| **Size** | 0 |

**dtype:** int64

In [8]:

```
#merge the data set
data = pd.merge(sales_data, features, on=["Store","Date","IsHoliday"])
data = pd.merge(data, store, on="Store")
```

In [9]:

```
# final check for missing values
data.isnull().sum()
```

Out [9]:

| | 0 |
|---|---|
| **Store** | 0 |
| **Dept** | 0 |
| **Date** | 0 |
| **Weekly_Sales** | 0 |
| **IsHoliday** | 0 |

| | 0 |
|---|---|
| **Temperature** | 0 |
| **Fuel_Price** | 0 |
| **MarkDown1** | 0 |
| **MarkDown2** | 0 |
| **MarkDown3** | 0 |
| **MarkDown4** | 0 |
| **MarkDown5** | 0 |
| **CPI** | 0 |
| **Unemployment** | 0 |
| **Type** | 0 |
| **Size** | 0 |

**dtype:** int64

In [10]:

```
# get shape of data set
data.shape
```

Out [10]: (421570, 16)

In [11]:

```
# get the info of data, number of row and column
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 421570 entries, 0 to 421569
Data columns (total 16 columns):
 #   Column        Non-Null Count   Dtype
---  ------        --------------   -----
 0   Store         421570 non-null  int64
 1   Dept          421570 non-null  int64
 2   Date          421570 non-null  object
 3   Weekly_Sales  421570 non-null  float64
 4   IsHoliday     421570 non-null  bool
 5   Temperature   421570 non-null  float64
 6   Fuel_Price    421570 non-null  float64
 7   MarkDown1     421570 non-null  float64
 8   MarkDown2     421570 non-null  float64
 9   MarkDown3     421570 non-null  float64
 10  MarkDown4     421570 non-null  float64
 11  MarkDown5     421570 non-null  float64
 12  CPI           421570 non-null  float64
 13  Unemployment  421570 non-null  float64
 14  Type          421570 non-null  object
 15  Size          421570 non-null  int64
dtypes: bool(1), float64(10), int64(3), object(2)
memory usage: 48.6+ MB
```

In [12]:

```
# get the descriptive statistics info
data.describe
```

Out [12]:

```
pandas.core.generic.NDFrame.describe
def describe(percentiles=None, include=None, exclude=None) -> Self

upper percentile is ``75``. The ``50`` percentile is the
same as the median.

For object data (e.g. strings or timestamps), the result's index
will include ``count``, ``unique``, ``top``, and ``freq``. The ``top``
is the most common value. The ``freq`` is the most common value's
frequency. Timestamps also include the ``first`` and ``last`` items.

If multiple object values have the highest count, then the
```

In [13]:

```
# checking the firstfew data
data.head()
```

Out [13]:

| | Store | Dept | Date | Weekly_Sales | IsHoliday | Temperature | Fuel_Price | MarkDown1 | MarkDown2 | MarkDown3 | MarkDown4 | MarkDown5 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 1 | 05/02/2010 | 24924.50 | False | 42.31 | 2.572 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 211 |
| **1** | 1 | 1 | 12/02/2010 | 46039.49 | True | 38.51 | 2.548 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 211 |
| **2** | 1 | 1 | 19/02/2010 | 41595.55 | False | 39.93 | 2.514 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 211 |
| **3** | 1 | 1 | 26/02/2010 | 19403.54 | False | 46.63 | 2.561 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 211 |
| **4** | 1 | 1 | 05/03/2010 | 21827.90 | False | 46.50 | 2.625 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 211 |

#Holiday and non holiday sales:

A comparison was made between holiday and non-holiday periods using bar plots.

In [14]:
```
# compare mean sales between holiday and non holiday
data.groupby("IsHoliday")["Weekly_Sales"].mean()
```

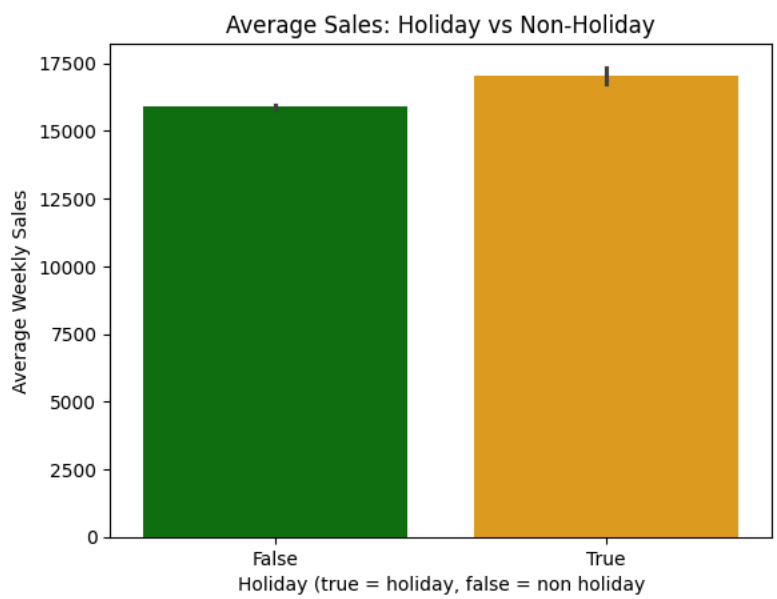
Out [14]:

| IsHoliday | Weekly_Sales |
|---|---|
| False | 15901.445069 |
| True | 17035.823187 |

**dtype:** float64

In [15]:
```
# visualization
sns.barplot(x="IsHoliday", y="Weekly_Sales", data=data, palette=["green", "orange"])
plt.title("Average Sales: Holiday vs Non-Holiday")
plt.xlabel("Holiday (true = holiday, false = non holiday")
plt.ylabel("Average Weekly Sales")

plt.show()
```

```
/tmp/ipykernel_311/3559090005.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set
`legend=False` for the same effect.

  sns.barplot(x="IsHoliday", y="Weekly_Sales", data=data, palette=["green", "orange"])
```




#Insight of above plot:

Holiday weeks tend to have higher average sales

Greater variability during holidays indicates spikes in demand

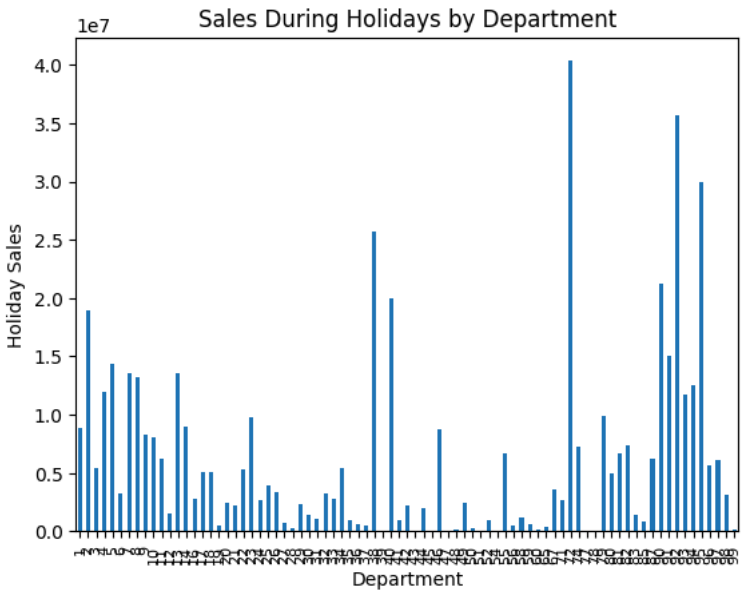
Confirms that holidays significantly impact revenue

In [16]:
```
# holiday sales by department
holiday_sales_dept = data[data["IsHoliday"] == True].groupby("Dept")["Weekly_Sales"].sum()

print(holiday_sales_dept)
```

```
Dept
1      8867168.77
2     18991468.92
3      5481134.64
4     12005820.97
5     14347362.90
         ...
95    29996233.40
96     5723180.51
97     6133824.50
98     3191644.19
99      119272.58
Name: Weekly_Sales, Length: 81, dtype: float64
```

In [17]:
```
#Visualization
```

```
holiday_sales_dept.plot(kind="bar")
plt.xlabel("Department")
plt.ylabel("Holiday Sales")
plt.xticks(rotation='vertical', size=8)
plt.title("Sales During Holidays by Department")
plt.show()
```




#Holiday sales in each department:

The plot is created to compare holiday revenue in each department

#Insights: -certain department produce high revenue during holiday weeks. -High revenue indicate that the department has better location and demanded product

In [18]:

```
# top 10 department with high holiday sales
top_holiday_sales_by_dept=holiday_sales_dept.sort_values(ascending=False)
```

In [19]:

```
top_holiday_sales_by_dept
```

Out [19]:

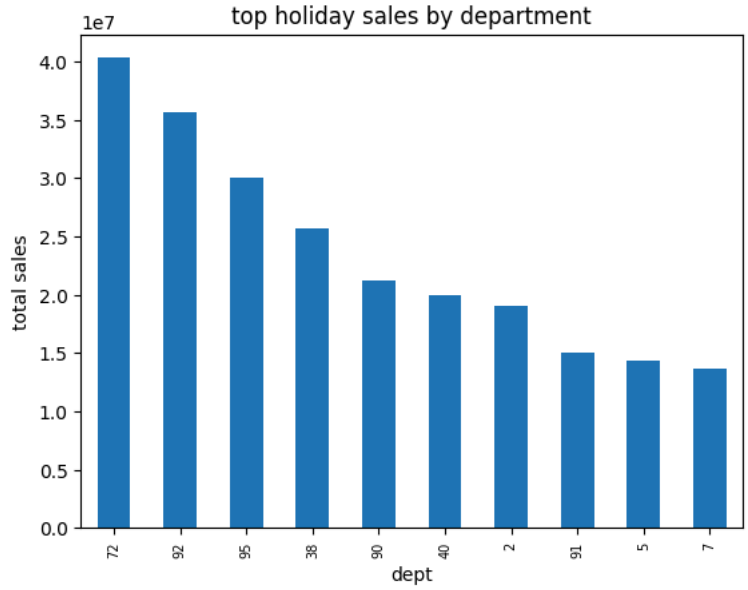
| | Weekly_Sales |
|---|---|
| **Dept** | |
| **72** | 40313605.46 |
| **92** | 35695397.10 |
| **95** | 29996233.40 |
| **38** | 25682245.57 |
| **90** | 21209128.22 |
| **...** | ... |
| **51** | 1711.58 |
| **47** | 250.93 |
| **78** | 169.76 |
| **39** | 13.41 |
| **43** | 0.25 |

81 rows × 1 columns

**dtype:** float64

In [20]:

```
top_holiday_sales_by_dept.head(10).plot(kind="bar")
plt.xlabel('dept')
plt.ylabel('total sales')
plt.title('top holiday sales by department')
plt.xticks(rotation='vertical', size=7)
plt.show()
```




In [21]:

```
# convert the date inform of month_year
data["month_year"] = pd.to_datetime(data["Date"], format="%d/%m/%Y").dt.strftime("%y-%m")
data["month_year"]
```

Out [21]:

| | month_year |
|---|---|
| **0** | 10-02 |
| **1** | 10-02 |
| **2** | 10-02 |
| **3** | 10-02 |
| **4** | 10-03 |
| **...** | ... |
| **421565** | 12-09 |
| **421566** | 12-10 |
| **421567** | 12-10 |
| **421568** | 12-10 |
| **421569** | 12-10 |

421570 rows × 1 columns

**dtype:** object

In [22]:

```
# the top month with the highest sales
data.groupby(["month_year"]).sum()["Weekly_Sales"].sort_values(ascending=False)
```

Out [22]:

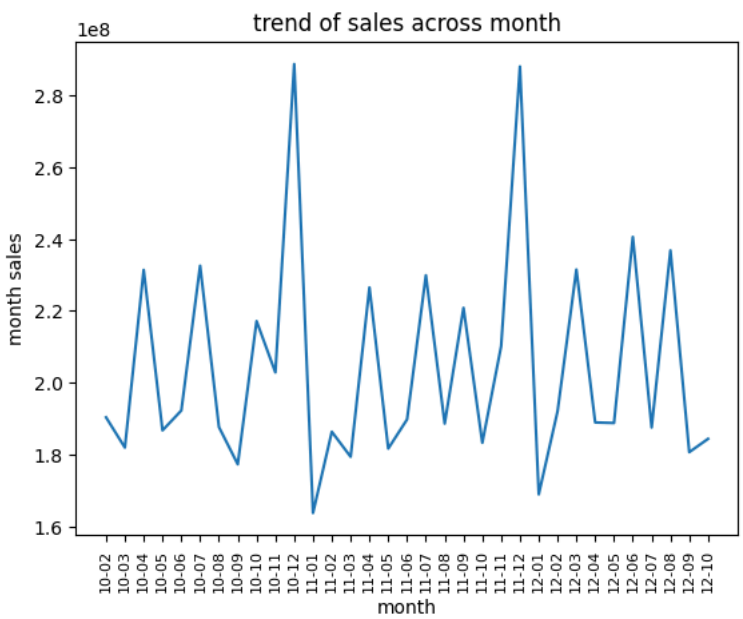
| month_year | Weekly_Sales |
|---|---|
| **10-12** | 2.887605e+08 |
| **11-12** | 2.880781e+08 |
| **12-06** | 2.406103e+08 |
| **12-08** | 2.368508e+08 |
| **10-07** | 2.325801e+08 |
| **12-03** | 2.315097e+08 |
| **10-04** | 2.314124e+08 |
| **11-07** | 2.299114e+08 |
| **11-04** | 2.265265e+08 |
| **11-09** | 2.208477e+08 |
| **10-10** | 2.171618e+08 |
| **11-11** | 2.101624e+08 |
| **10-11** | 2.028534e+08 |
| **10-06** | 1.922462e+08 |

| month_year | Weekly_Sales |
|---|---|
| **12-02** | 1.920636e+08 |
| **10-02** | 1.903330e+08 |
| **11-06** | 1.897734e+08 |
| **12-04** | 1.889209e+08 |
| **12-05** | 1.887665e+08 |
| **11-08** | 1.885993e+08 |
| **10-08** | 1.876401e+08 |
| **12-07** | 1.875095e+08 |
| **10-05** | 1.867109e+08 |
| **11-02** | 1.863313e+08 |
| **12-10** | 1.843617e+08 |
| **11-10** | 1.832613e+08 |
| **10-03** | 1.819198e+08 |
| **11-05** | 1.816482e+08 |
| **12-09** | 1.806455e+08 |
| **11-03** | 1.793564e+08 |
| **10-09** | 1.772679e+08 |
| **12-01** | 1.688945e+08 |
| **11-01** | 1.637040e+08 |

**dtype:** float64

In [23]:

```
# the trend of sales
sales_trend = data.groupby(["month_year"]).sum()["Weekly_Sales"]
plt.plot(sales_trend.index, sales_trend.values)
plt.ylabel('month sales')
plt.xlabel('month')
plt.title('trend of sales across month')
plt.xticks(rotation='vertical', size=8)
plt.show()
```




#Sales trend over time:

A time-series analysis was performed to observe how sales change over time.

#Insight:

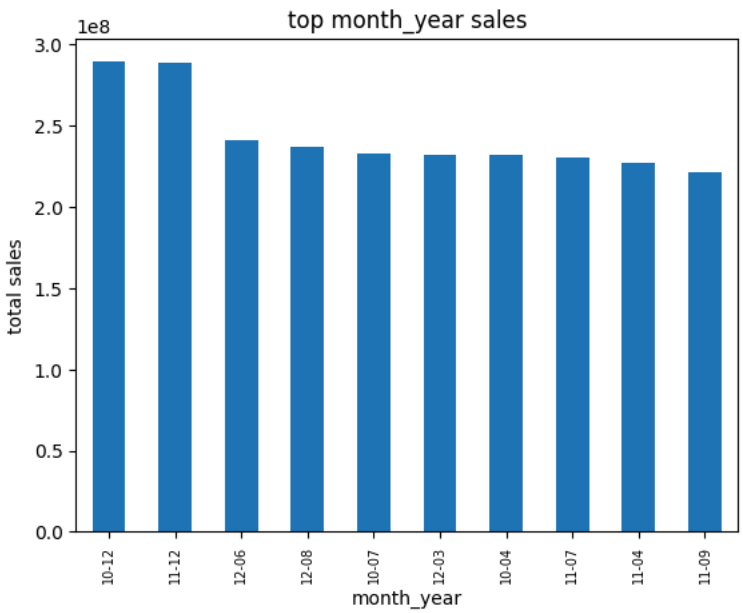
Sales show fluctuations across different periods

Peaks are observed during certain weeks, likely corresponding to holidays

Indicates seasonal patterns in customer purchasing behavior

In [24]:

```
# the top 10 sales by month
sales_trend = data.groupby(["month_year"]).sum()["Weekly_Sales"].sort_values(ascending =False)
sales_trend.head(10).plot(kind="bar")
sales_trend.head(10).plot(kind="bar")
plt.xlabel('month_year')
plt.ylabel('total sales')
plt.title('top month_year sales')
plt.xticks(rotation='vertical', size=7)
plt.show()
```



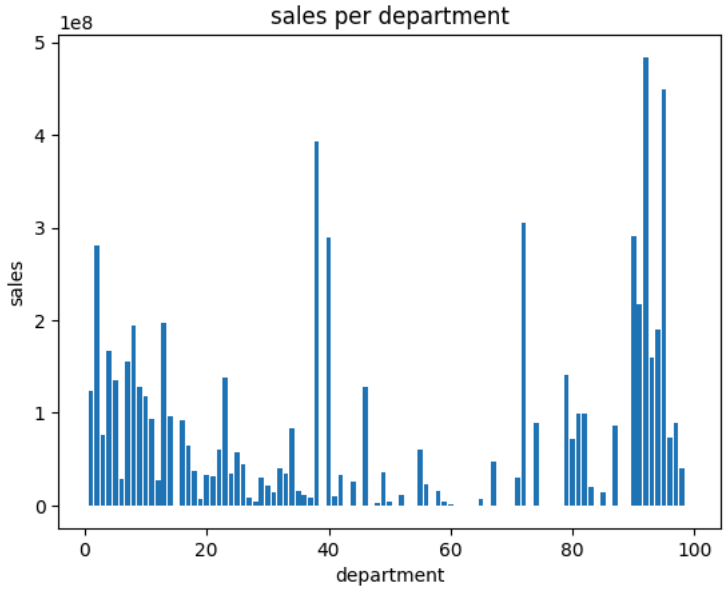

In [26]:

```
#overall sales by department
mostsold_dept=data.groupby(["Dept"]).sum()["Weekly_Sales"].sort_values(ascending=False)
print(mostsold_dept)
```

```
Dept
92    4.839433e+08
95    4.493202e+08
38    3.931181e+08
72    3.057252e+08
90    2.910685e+08
         ...
51    3.057283e+04
78    1.714710e+03
39    1.779800e+02
43    1.432000e+01
47   -4.962930e+03
Name: Weekly_Sales, Length: 81, dtype: float64
```

In [27]:

```
plt.bar(mostsold_dept.index, mostsold_dept.values)
plt.ylabel('sales')
plt.xlabel('department')
plt.title('sales per department')
plt.show()
```




The plot show that certain department contribute more revenue

In [28]:
```
data["Store"].max()
```

Out [28]: 45

In [29]:
```
data["Store"].min()
```

Out [29]: 1

In [30]:
```
data["Store"]
```

Out [30]:

| | Store |
|---|---|
| **0** | 1 |
| **1** | 1 |
| **2** | 1 |
| **3** | 1 |
| **4** | 1 |
| **...** | ... |
| **421565** | 45 |
| **421566** | 45 |
| **421567** | 45 |
| **421568** | 45 |
| **421569** | 45 |

421570 rows × 1 columns

**dtype:** int64

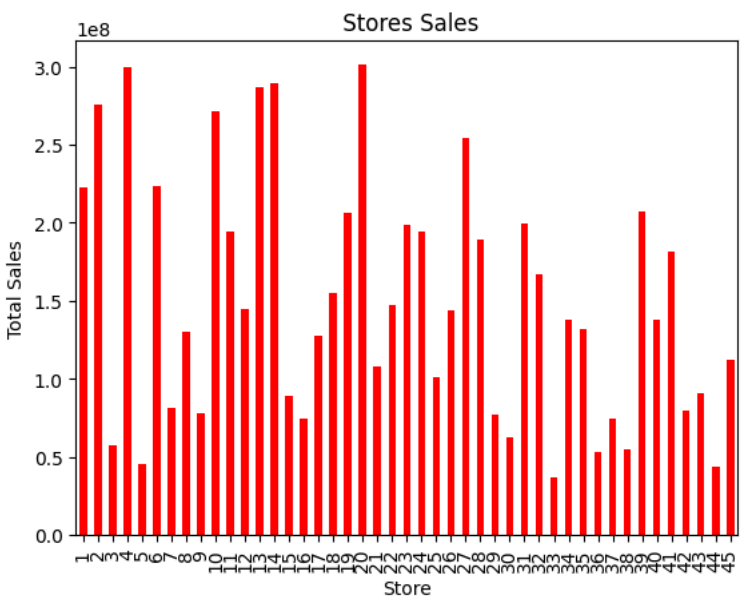
In [31]:
```
store_sales=data.groupby(["Store"]).sum()["Weekly_Sales"]
print(store_sales)
```

```
Store
1     2.224028e+08
2     2.753824e+08
3     5.758674e+07
4     2.995440e+08
5     4.547569e+07
6     2.237561e+08
7     8.159828e+07
8     1.299512e+08
9     7.778922e+07
10    2.716177e+08
11    1.939628e+08
12    1.442872e+08
13    2.865177e+08
14    2.889999e+08
15    8.913368e+07
16    7.425243e+07
17    1.277821e+08
18    1.551147e+08
19    2.066349e+08
20    3.013978e+08
21    1.081179e+08
22    1.470756e+08
23    1.987506e+08
24    1.940160e+08
```

```
25    1.010612e+08
26    1.434164e+08
27    2.538559e+08
28    1.892637e+08
29    7.714155e+07
30    6.271689e+07
31    1.996139e+08
32    1.668192e+08
33    3.716022e+07
34    1.382498e+08
35    1.315207e+08
36    5.341221e+07
37    7.420274e+07
38    5.515963e+07
39    2.074455e+08
40    1.378703e+08
41    1.813419e+08
42    7.956575e+07
43    9.056544e+07
44    4.329309e+07
45    1.123953e+08
Name: Weekly_Sales, dtype: float64
```

In [32]:

```
store_sales.plot(kind="bar", color ="red")
plt.xlabel("Store")
plt.ylabel("Total Sales")
plt.title(" Stores Sales")
plt.show()
```




In [33]:

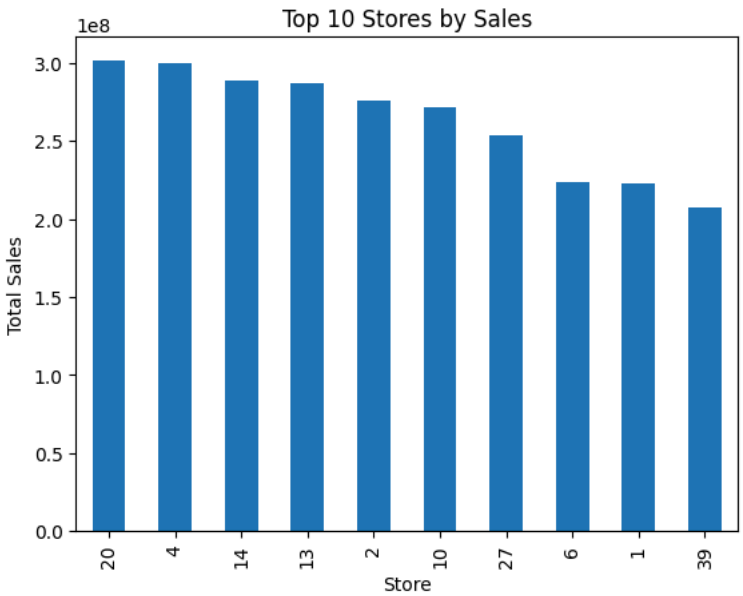
```
# top performing store by sales
top_store=data.groupby(["Store"]).sum()["Weekly_Sales"].sort_values(ascending=False)
print(top_store)
```

```
Store
20    3.013978e+08
4     2.995440e+08
14    2.889999e+08
13    2.865177e+08
2     2.753824e+08
10    2.716177e+08
27    2.538559e+08
6     2.237561e+08
1     2.224028e+08
39    2.074455e+08
19    2.066349e+08
31    1.996139e+08
23    1.987506e+08
24    1.940160e+08
11    1.939628e+08
28    1.892637e+08
41    1.813419e+08
32    1.668192e+08
18    1.551147e+08
22    1.470756e+08
12    1.442872e+08
26    1.434164e+08
34    1.382498e+08
40    1.378703e+08
35    1.315207e+08
8     1.299512e+08
17    1.277821e+08
45    1.123953e+08
21    1.081179e+08
25    1.010612e+08
43    9.056544e+07
15    8.913368e+07
7     8.159828e+07
42    7.956575e+07
9     7.778922e+07
29    7.714155e+07
```

```
16    7.425243e+07
37    7.420274e+07
30    6.271689e+07
3     5.758674e+07
38    5.515963e+07
36    5.341221e+07
5     4.547569e+07
44    4.329309e+07
33    3.716022e+07
Name: Weekly_Sales, dtype: float64
```

In [34]:

```
#visualization of the top 10
top_store.head(10).plot(kind="bar")
plt.xlabel("Store")
plt.ylabel("Total Sales")
plt.title("Top 10 Stores by Sales")
plt.show()
```




In [35]:

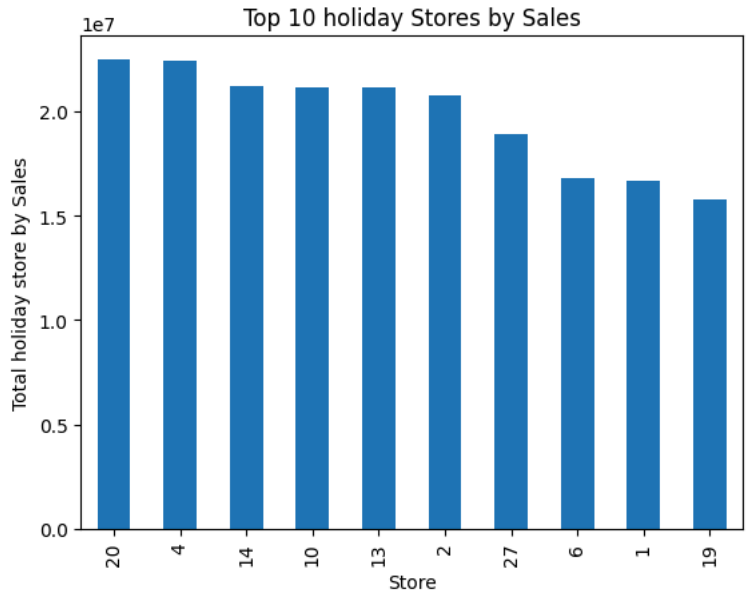
```
#Store with high holiday sales
holiday_store_sales = data[data["IsHoliday"]==True].groupby("Store")["Weekly_Sales"].sum().sort_values(ascendi

print(holiday_store_sales.sort_values(ascending=False))
```

```
Store
20    22490350.81
4     22431026.24
14    21205829.98
10    21137559.49
13    21130438.06
2     20792669.00
27    18922992.78
6     16809079.27
1     16657476.56
19    15770467.34
39    15511274.80
31    15000260.30
28    14782446.05
24    14750982.51
23    14625422.94
11    14483944.85
41    13349478.56
32    12037840.83
18    11694221.61
12    11381404.20
22    10848746.56
35    10743484.57
26    10720468.49
34    10419780.89
40    10080340.75
17     9797969.71
8      9753308.60
45     8362937.13
21     8264913.09
25     7396768.42
15     7064060.18
7      6724002.65
43     6359462.78
29     6069578.89
9      5889508.21
42     5676941.58
16     5667336.46
37     5075250.50
3      4378110.50
30     4368593.07
38     3815098.78
36     3676406.30
5      3595016.07
44     2960356.01
```

```
33     2625945.19
Name: Weekly_Sales, dtype: float64
```

In [36]:

```
holiday_store_sales.head(10).plot(kind="bar")
plt.xlabel("Store")
plt.ylabel("Total holiday store by Sales")
plt.title("Top 10 holiday Stores by Sales")
plt.show()
```




#Top Performing Stores

Sales were grouped by store to identify the highest-performing locations.

#Insight:

Certain stores (e.g., Store 20) generate significantly higher revenue

Indicates variation in performance across locations

Suggests influence of store size, location, or customer demand

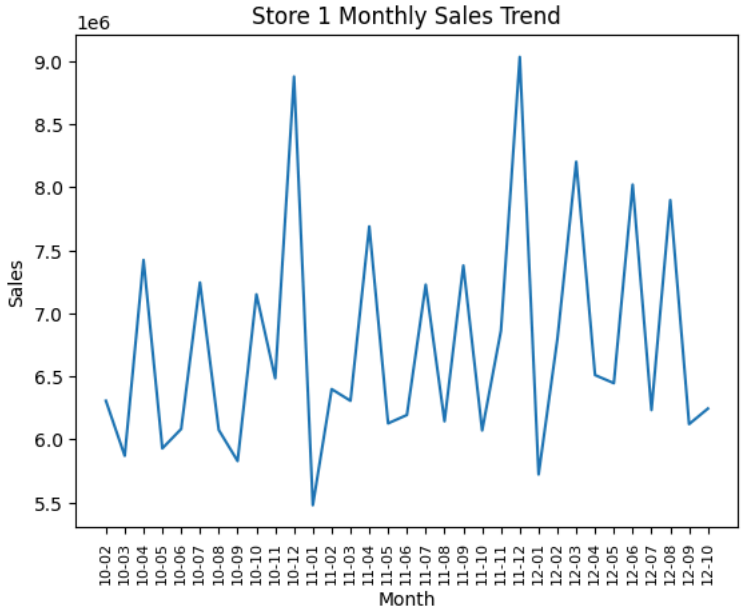
In [37]:

```
#monthly sales for each store
store_trend = data.groupby(["month_year","Store"])["Weekly_Sales"].sum()

print(store_trend)
```

```
month_year  Store
10-02       1        6307344.10
            2        8264347.77
            3        1711198.23
            4        8299040.36
            5        1202728.00
                        ...
12-10       41       5452445.75
            42       2261705.49
            43       2473507.39
            44       1360020.41
            45       2946326.39
Name: Weekly_Sales, Length: 1485, dtype: float64
```

In [38]:

```
# trend in monthly sales in store 1
store1 = data[data["Store"] == 1].groupby("month_year")["Weekly_Sales"].sum()

plt.plot(store1.index, store1.values)
plt.xlabel("Month")
plt.ylabel("Sales")
plt.xticks(rotation='vertical', size=8)
plt.title("Store 1 Monthly Sales Trend")
plt.show()
```




In [39]:

```
# department in store 20 with hi revenue
store20 = data[data["Store"] == 20]
print(store20)
```

```
        Store  Dept        Date  Weekly_Sales  IsHoliday  Temperature  \
187335     20     1  05/02/2010      46021.21      False        25.92
187336     20     1  12/02/2010      52734.17       True        22.12
187337     20     1  19/02/2010      55649.79      False        25.43
187338     20     1  26/02/2010      36805.42      False        32.32
187339     20     1  05/03/2010      41206.78      False        31.75
...       ...   ...         ...          ...        ...          ...
197544     20    99  07/09/2012          0.67       True        76.36
197545     20    99  14/09/2012          0.05      False        64.84
197546     20    99  21/09/2012         29.91      False        60.94
197547     20    99  05/10/2012        660.00      False        60.77
197548     20    99  12/10/2012        200.00      False        47.20

        Fuel_Price  MarkDown1  MarkDown2  MarkDown3  MarkDown4  MarkDown5  \
187335       2.784       0.00       0.00       0.00       0.00       0.00
187336       2.773       0.00       0.00       0.00       0.00       0.00
187337       2.745       0.00       0.00       0.00       0.00       0.00
187338       2.754       0.00       0.00       0.00       0.00       0.00
187339       2.777       0.00       0.00       0.00       0.00       0.00
...            ...        ...        ...        ...        ...        ...
197544       3.911   11783.56      -0.99      83.85    1938.40    3708.67
197545       3.948   10404.97       0.00      40.11    1481.31    8655.59
197546       4.038    8475.15     168.56      49.78    1727.92    9072.77
197547       3.985    5800.66       0.00      33.68    4286.15    3224.25
197548       4.000    2449.31       0.00      35.08    1019.90    9743.56

               CPI  Unemployment Type    Size month_year
187335  204.247194         8.187    A  203742      10-02
187336  204.385747         8.187    A  203742      10-02
187337  204.432100         8.187    A  203742      10-02
187338  204.463087         8.187    A  203742      10-02
187339  204.494073         8.187    A  203742      10-03
...            ...           ...  ...     ...        ...
197544  215.218957         7.280    A  203742      12-09
197545  215.358376         7.280    A  203742      12-09
197546  215.547546         7.280    A  203742      12-09
197547  215.925886         7.293    A  203742      12-10
197548  216.115057         7.293    A  203742      12-10

[10214 rows x 17 columns]
```

In [40]:

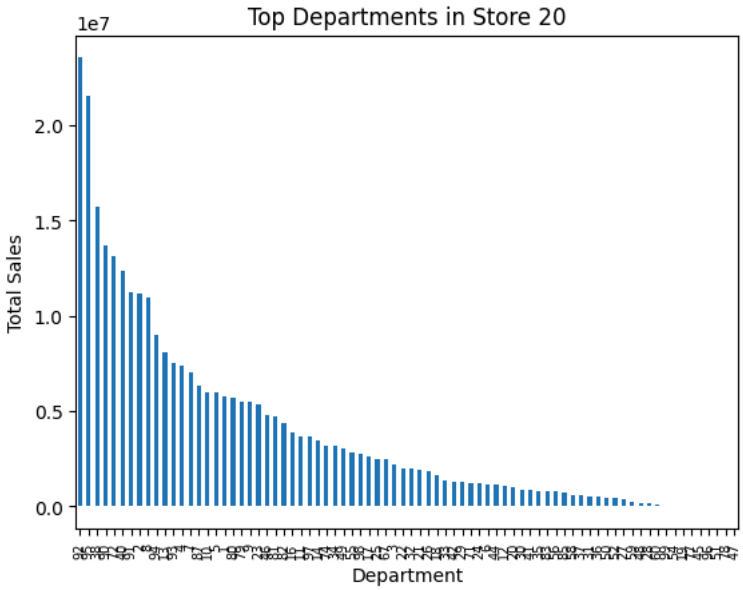
```
#the department in store 20 wich have high revenue
dept_sales = store20.groupby("Dept")["Weekly_Sales"].sum().sort_values(ascending=False)
print(dept_sales)
```

```
Dept
92    23542625.04
95    21537795.62
38    15759345.93
90    13707777.99
72    13154550.44
         ...
45         666.18
96         244.27
51         161.18
78          29.00
47        -378.58
Name: Weekly_Sales, Length: 78, dtype: float64
```

In [41]:

```
# department in store 20 wich generate high sales
dept_sales.plot(kind="bar")
```

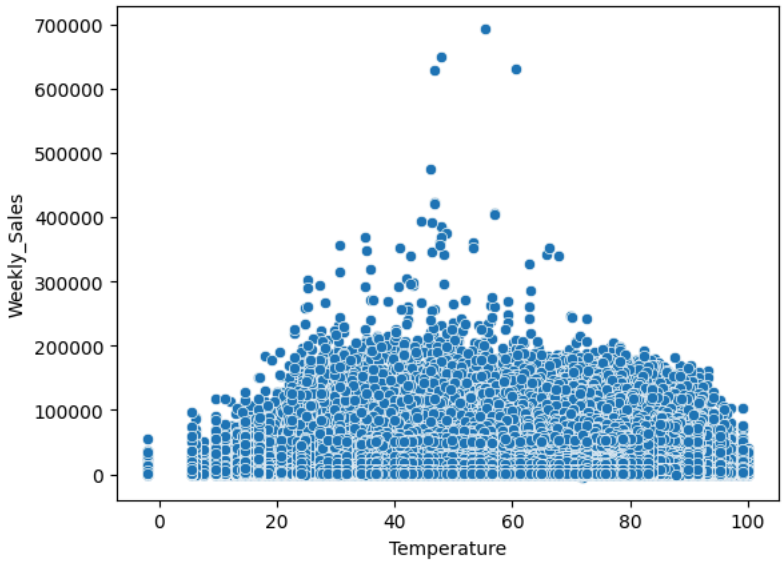
```
plt.title("Top Departments in Store 20")
plt.xlabel("Department")
plt.ylabel("Total Sales")
plt.xticks(size=7)
plt.show()
```




In [42]:

```
# sales trend by temperature
sns.scatterplot(x="Temperature", y="Weekly_Sales", data=data)
```

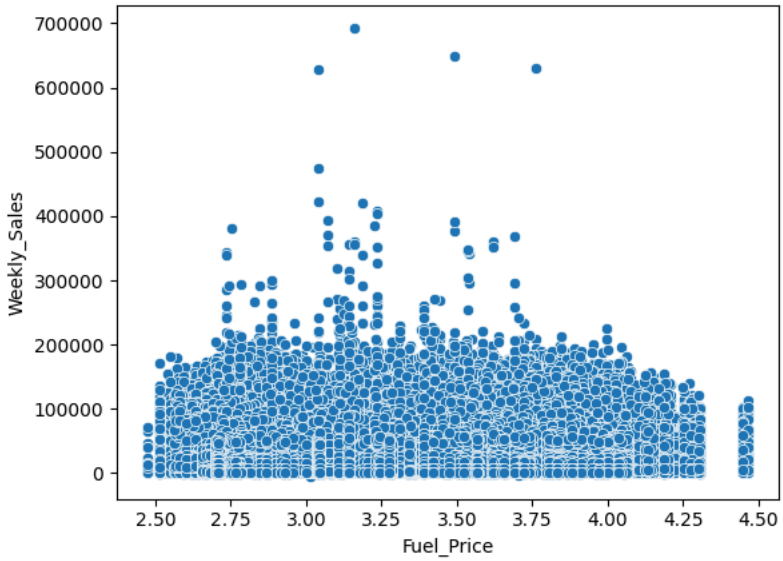
Out [42]: <Axes: xlabel='Temperature', ylabel='Weekly_Sales'>




In [43]:

```
# trend of sales by fuel price
sns.scatterplot(x="Fuel_Price", y="Weekly_Sales", data=data)
```

Out [43]: <Axes: xlabel='Fuel_Price', ylabel='Weekly_Sales'>




#Fuel Price vs Sales:

A scatterplot was used to analyze the relationship between fuel price and weekly sales.

#Insight:

Data points are widely scattered with no clear pattern

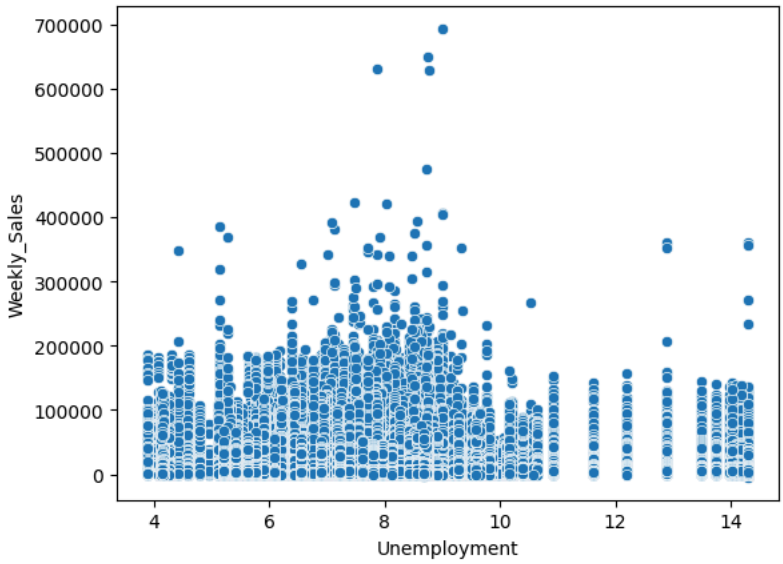
Indicates a weak or no relationship

Suggests fuel price does not significantly affect sales

In [44]:

```
# impact of unemployment on sales
sns.scatterplot(x="Unemployment", y="Weekly_Sales", data=data)
```

Out [44]: <Axes: xlabel='Unemployment', ylabel='Weekly_Sales'>




#Unemployment vs Sales:

The relationship between unemployment rate and sales was examined.

#Insight:

Slight negative trend observed

Higher unemployment may slightly reduce sales

However, the relationship is weak and not a major driver

In [45]:

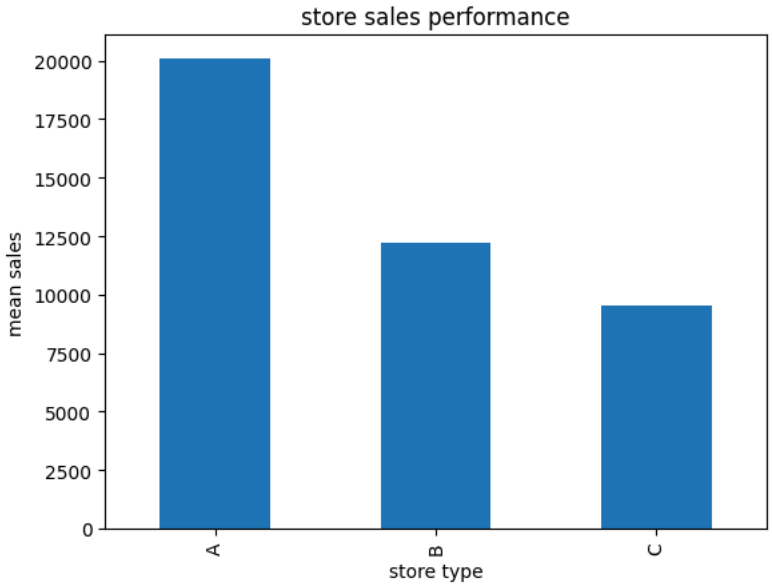
```
# sales performance by store type
store_type_performance=data.groupby("Type")["Weekly_Sales"].mean()
print (store_type_performance)
```

```
Type
A    20099.568043
B    12237.075977
```

```
C     9519.532538
Name: Weekly_Sales, dtype: float64
```

In [46]:

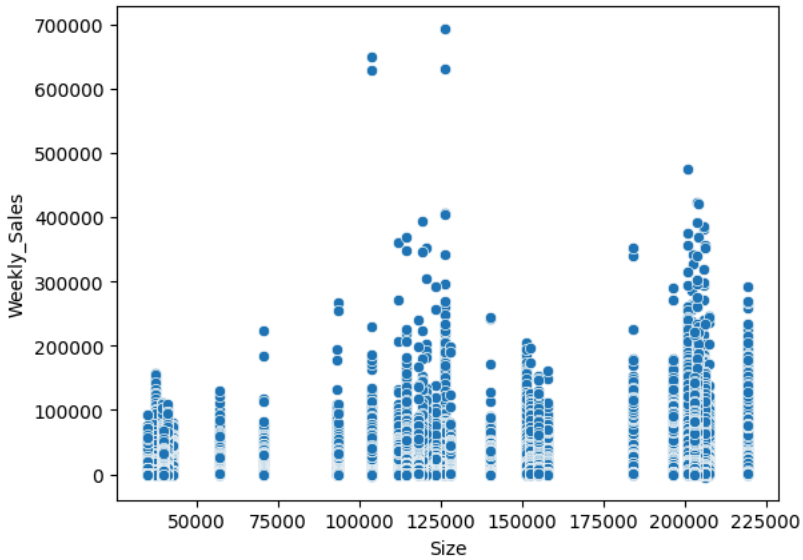
```
# visualize store type performance
store_type_performance.plot(kind="bar")
plt.xlabel('store type')
plt.ylabel('mean sales')
plt.title('store sales performance')
plt.show()
```




In [47]:

```
#sales by store size
sns.scatterplot(x="Size", y="Weekly_Sales", data=data)
```

Out [47]: <Axes: xlabel='Size', ylabel='Weekly_Sales'>




#Store Size vs Sales:

A scatterplot was used to analyze the effect of store size on sales.

#Insight:

A slight positive relationship is observed

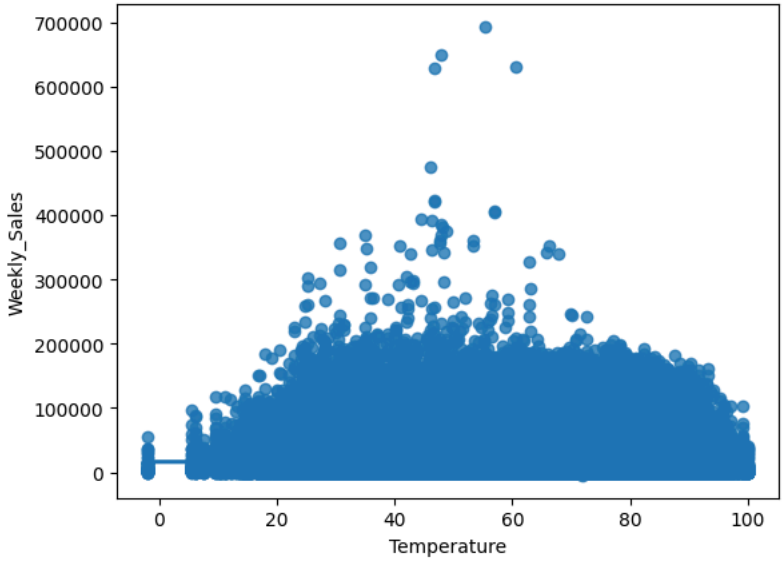
Larger stores tend to generate higher sales

However, variability suggests other factors also influence performance

In [49]:

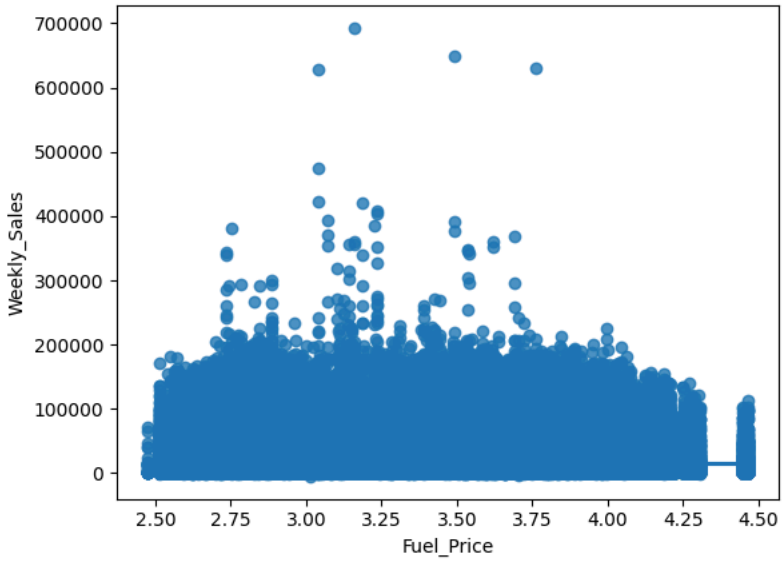
```
#clear view the plot to see how the dot are organized
sns.regplot(x="Temperature", y="Weekly_Sales", data=data)
```

Out [49]: <Axes: xlabel='Temperature', ylabel='Weekly_Sales'>



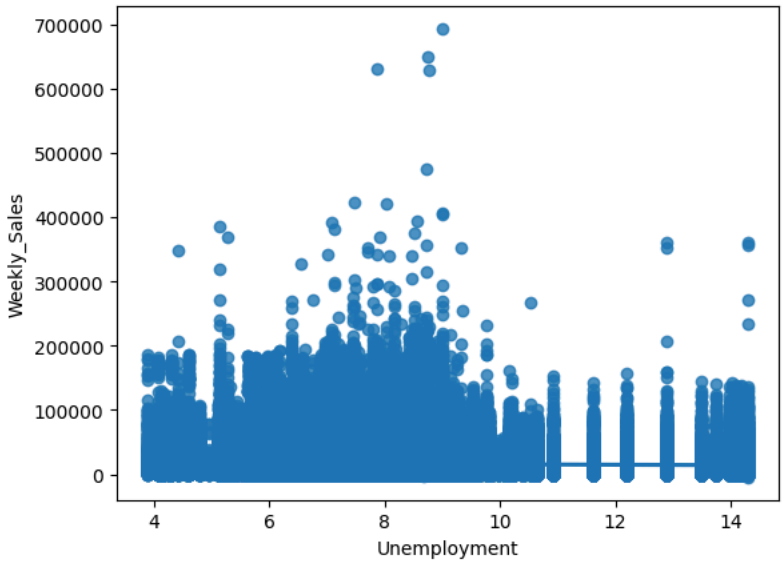

In [50]:
```
sns.regplot(x="Fuel_Price", y="Weekly_Sales", data=data)
```

Out [50]: <Axes: xlabel='Fuel_Price', ylabel='Weekly_Sales'>



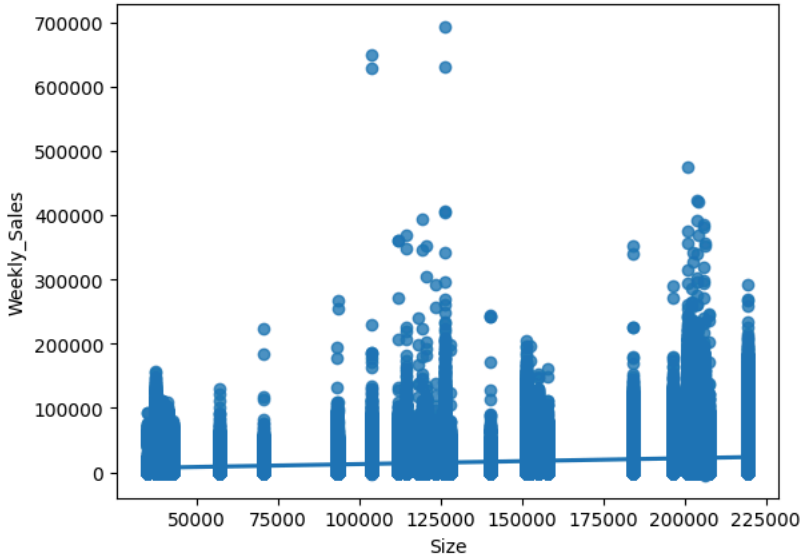

In [51]:
```
sns.regplot(x="Unemployment", y="Weekly_Sales", data=data)
```

Out [51]: <Axes: xlabel='Unemployment', ylabel='Weekly_Sales'>



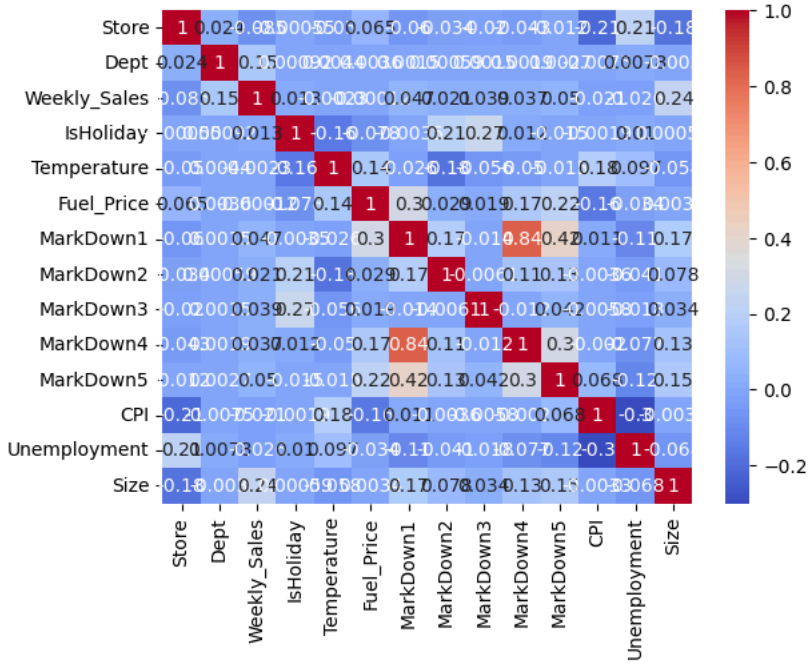

In [52]:
```
sns.regplot(x="Size", y="Weekly_Sales", data=data)
```

Out [52]: <Axes: xlabel='Size', ylabel='Weekly_Sales'>




In [53]:

```
#relationsgip between weekly sales and other external factors
corr = data.corr(numeric_only=True )

sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.show()
```




The heatmap show that External economic and environmental factors have minimal impact on sales

In [54]:

```
# total promotion effect
data["Total_MarkDown"] = data[["MarkDown1","MarkDown2","MarkDown3","MarkDown4","MarkDown5"]].sum(axis=1)
data["Total_MarkDown"]
```

Out [54]:

| | Total_MarkDown |
|---|---|
| **0** | 0.00 |
| **1** | 0.00 |
| **2** | 0.00 |
| **3** | 0.00 |
| **4** | 0.00 |
| **...** | ... |
| **421565** | 9468.01 |
| **421566** | 9659.00 |
| **421567** | 6554.03 |

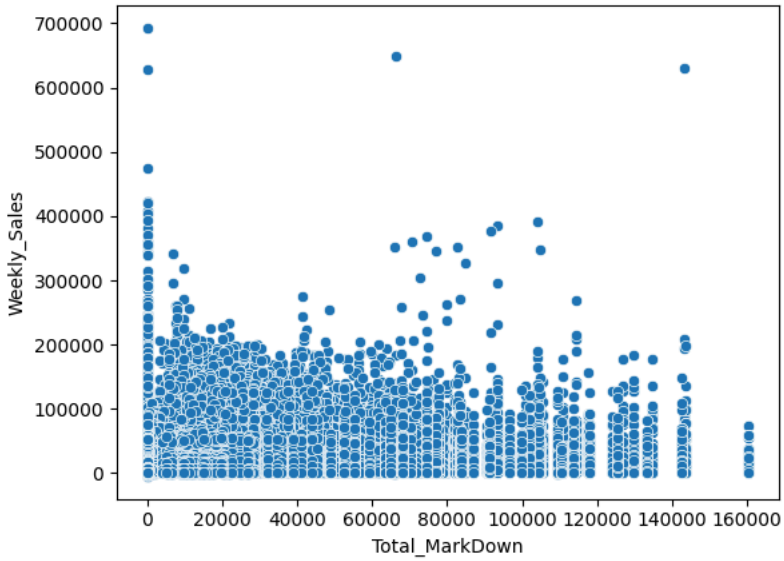
| | Total_MarkDown |
|---|---|
| **421568** | 3982.42 |
| **421569** | 5247.26 |

421570 rows × 1 columns

**dtype:** float64

In [55]:
```
#relationship between promotion distribution and sales
sns.scatterplot(x="Total_MarkDown", y="Weekly_Sales", data=data)
```

Out [55]: <Axes: xlabel='Total_MarkDown', ylabel='Weekly_Sales'>




#Promotions (MarkDown) Analysis:

Promotional data (MarkDown columns) were combined to evaluate their effect on sales.

#Insight:

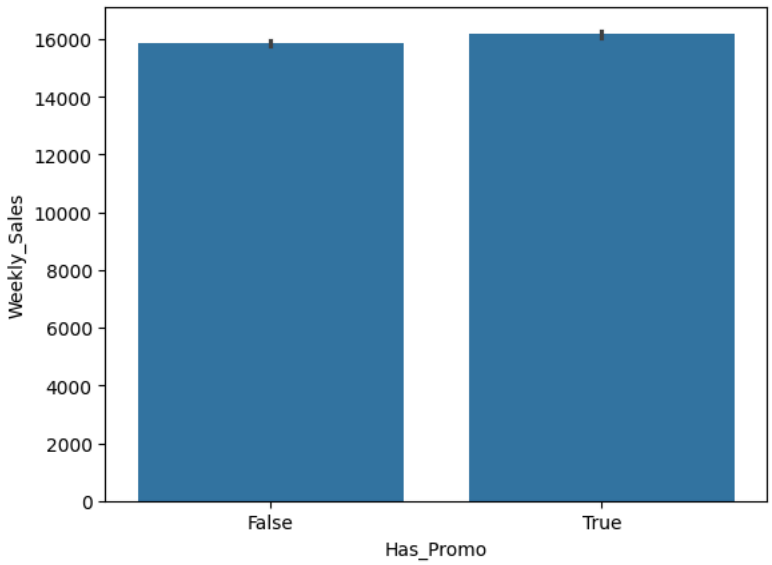
Promotions show a positive relationship with sales

Stores with promotions tend to perform better

Indicates that discount strategies are effective in boosting revenue

In [57]:
```
# store with promotions and stores with no promotions
data["Has_Promo"] = data["Total_MarkDown"] > 0

sns.barplot(x="Has_Promo", y="Weekly_Sales", data=data)
```

Out [57]: <Axes: xlabel='Has_Promo', ylabel='Weekly_Sales'>




In [58]:
```
#correlation
data[["Weekly_Sales","Total_MarkDown"]].corr()
```

Out [58]:

| | Weekly_Sales | Total_MarkDown |
|---|---|---|
| **Weekly_Sales** | 1.000000 | 0.065237 |
| **Total_MarkDown** | 0.065237 | 1.000000 |

#Conclusion:

This analysis demonstrates that internal business strategies such as promotions and holiday planning have a stronger impact on sales compared to external economic factors. Retailers can improve performance by focusing on promotional campaigns and optimizing store operations.